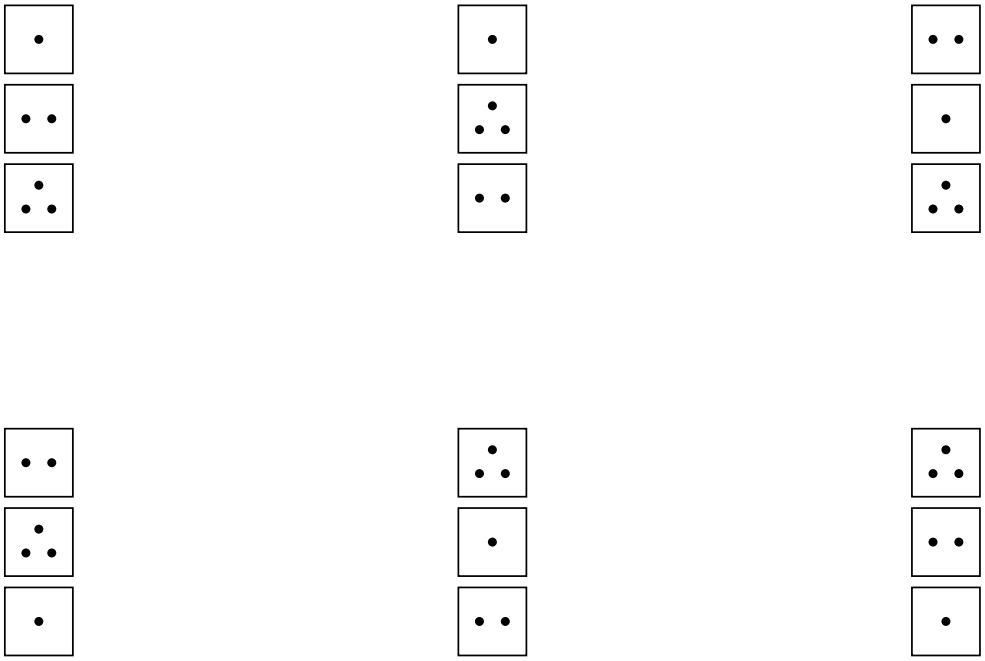
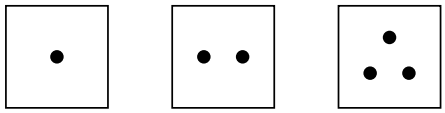
Cards move left to right. At each bar, look at the two cards at its ends: the one with fewer dots goes up. Equal cards stay. The bars stay the same for every start. The machine sorts when the fewest dots finish at the top.

Problem 1: Try every start. Circle the starts that finish in order.





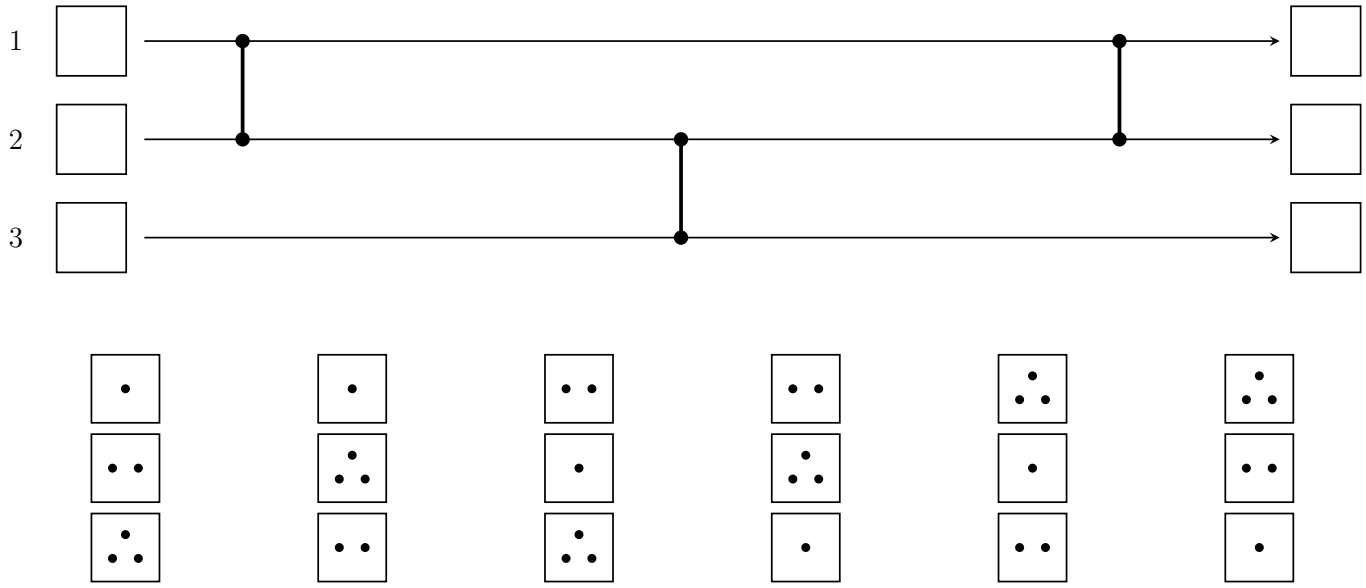
Problem 2: Build a machine that sorts these three cards from every start. Use as few bars as you can.

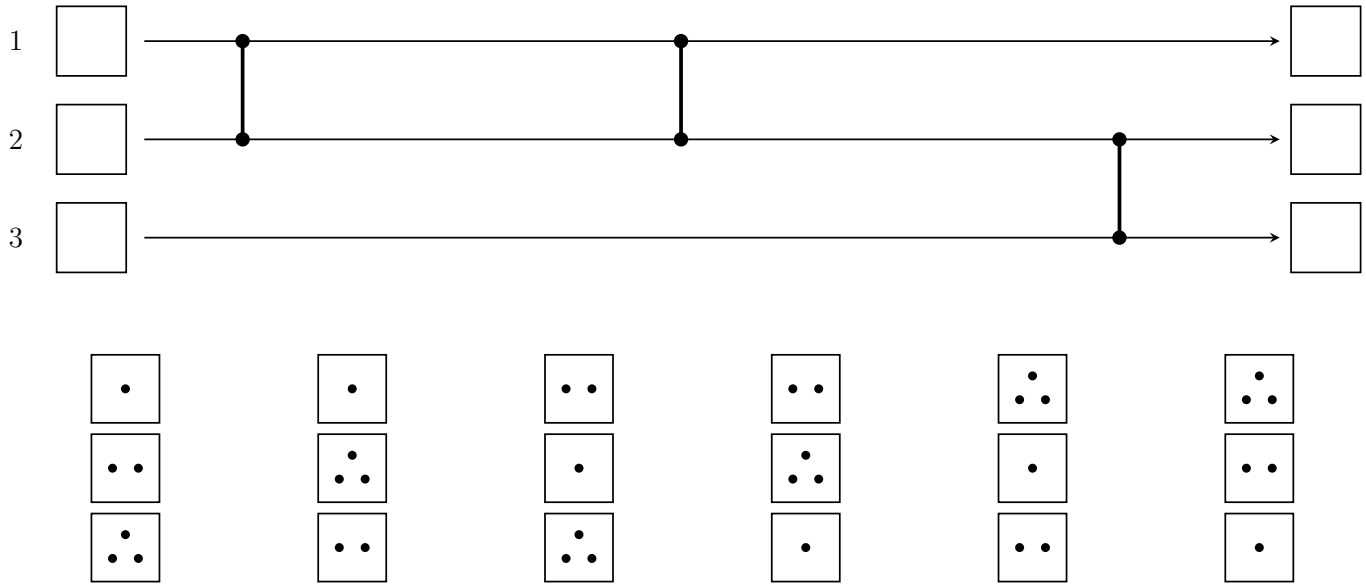




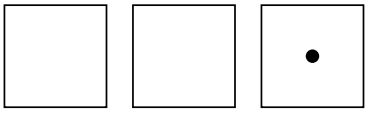


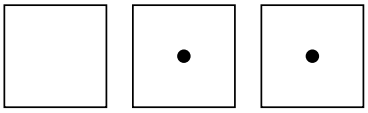
Problem 3: For each machine, circle every start that finishes in the wrong order.





Problem 4: Build a machine that always sorts the left set but sometimes fails on the right set. Make another that always sorts the right set but sometimes fails on the left set.

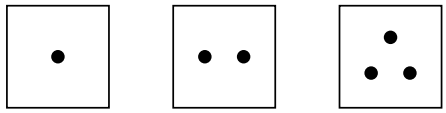


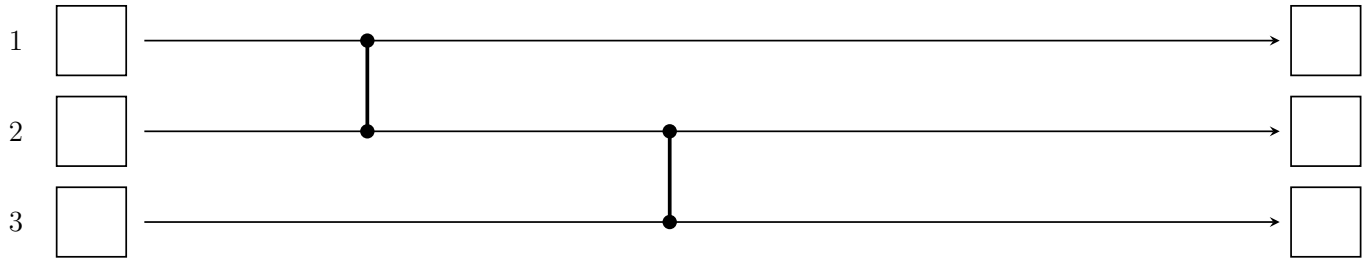


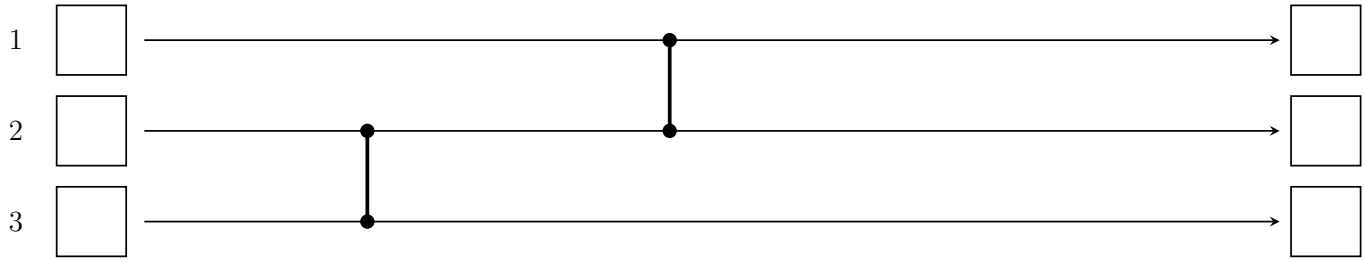


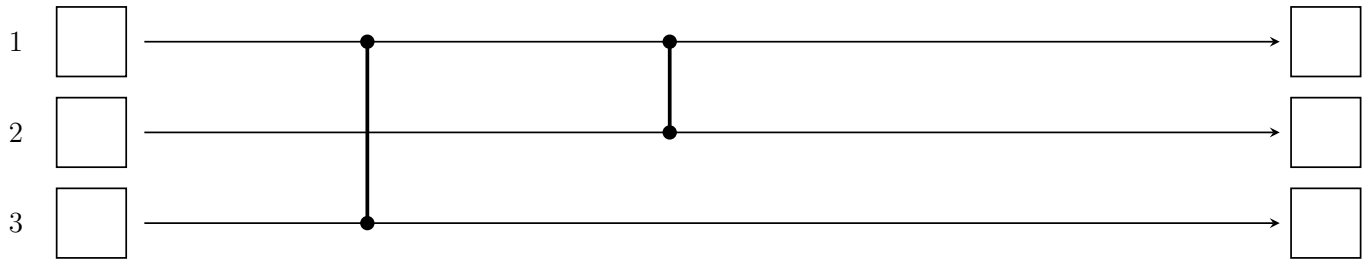


Problem 5: Each machine needs one more bar at the right. Add it so that every start of the three cards finishes in order.









Problem 6: Build a machine that sorts these four cards from every start. Use as few bars as you can.

